

Composite single neural PID controller based on fuzzy self-tuning gain and RBF network identification

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Abstract: Nonlinear, time-varying and uncertainty matter is an important issue in order to improve conventional PID control technology due to the rapid development of control technologies for a wide range of industries. Artificial neural PID controller such as the single neural PID controlling is a current interesting research topic because of its potential ability of auto-tuning by simulating the human brain functions. In the conventional single neural network control method, it is difficult to realize a high learning speed and online tuning PID gains. A new approach to improve the single neuron PID control technology is to combine fuzzy controlling with RBF neural network. Such a composite neuron adaptive PID controller with fuzzy self-tuning gain takes advantages of online identification and self-tuning gain thus to speed up the learning process and rationally adjust the gain coefficient, providing a much improved solution to reduce overshoot and to speed up self-learning. In this article, we show our two simulations including a second-order motion control and time-delay model to verify the effectiveness of the composite neuron adaptive PID controller method. We conclude that the adaptive fuzzy gain compound neural network PID control is superior in controlling nonlinear and time-delay objects over the conventional method. The improved control algorithm has excellent tracking performance and accuracy to meet the requirements for the next generation of intelligent controllers.

Key Words: Fuzzy auto-tuning gain, RBF network identification, Single neural PID controller

1 Introduction

PID control with its simple structure, easy to implement and good robustness is one of the most commonly used method in industrial control. However PID controller is not effective in the control of complex systems such as nonlinear, time delay because of losing parameter online adjusting function. Neural network has been widely used in the field of pattern recognition and control by simulating the human brain neurons which realizes the function of self-adaptive and self-learning. In recent years, many neurons control algorithm had been proposed, such as single neuron PID control and BP neural network, etc.^[1].

The Single neuron PID control received the great attention by people because it not only keeps the simple structure and easy to implement but also can tune parameters online, the model of single neuron is shown in figure 1. To get a good control effect, the conventional PID always need to create an accurately modeling of controlled object, but the single neuron control can avoid the complex modeling process by tuning parameters online^[2]. However there are still some deficiencies in single neuron controller like the weights need a long time to adjust, the learning process may step into local convergence and the sensitive coefficient K of the control system can not self-tuning whose value will directly impact on the control effect of the whole control system^[3]. So the single neuron PID controller needs to be improved^[4].

In order to overcome the weakness of the single neuron PID controller, a new approach has been studied which combining RBF neural network identification and single neuron PID controller together to get a compound neural network controller, at the same time, the fuzzy logic controller has been used to auto-tuning the K value for an optimal control effect. The improved controller take the advantage of the excellent complementarities between fuzzy control and neural network, using RBF to identify the controlled object, which can not only speed up the learning process and the convergence of the controller but also avoid the local convergence during the learning process; At the same time, using fuzzy logic controller^[5] to use the input information more efficient and adjust the sensitive coefficient K of the single neuron PID reasonably, improve the dynamic performance and robustness of the controller, reduce the overshoot effectively^[6].

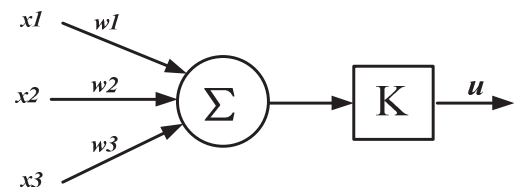


Fig 1. The single neuron model structure

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2 Design of the improved control system based on fuzzy gain auto-tuning and RBF network identification

The single neuron PID controller not only keeps the simple structure of conventional PID controller but also can online tune its parameters by neural network auto-learning. However it still can not self-tuning gain coefficient K and the learning process of weight is not fast enough^[8]. The proposed fuzzy auto-tuning gain composite neural network PID controller combining the RBF neural network and fuzzy control to improve the single neuron PID controller, the structure of the improved control system is shown in figure 2.

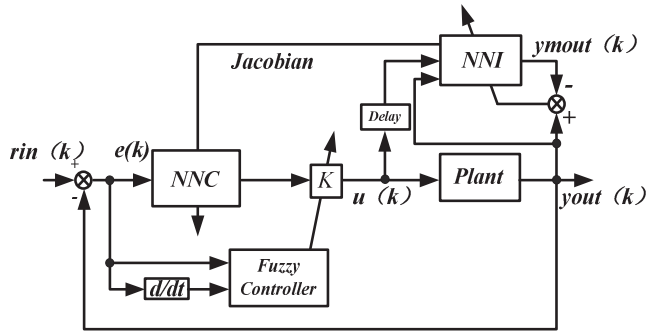


Fig 2: The structure of fuzzy auto-tuning gain composite neural network PID control system

NNC shows in figure 2 represent the single neuron adaptive PID controller, it completes the online tuning of proportion, integral and differential parameters; NNI represent RBF neural network, it will transport the controlled object (Plant)'s Jacobian information to NNC to speed up the learning process and reduce error; Fuzzy controller completes the online tuning of sensitive coefficient K according to the fuzzy reasoning rules to reduce the overshoot, improve the dynamic performance.

2.1 RBF neural network online identification

The low speed of weights learning process in single neuron PID controller will cause lag issues^[8], and RBF neural network can approximate nonlinear function in arbitrary precision to achieve a global optimal, so combine them together can effectively accelerate the learning process of single neuron PID controller. Figure 3 is the structure of RBF neural network topology, and the Gauss nuclear function has been used in this article as hidden layer activation function which is simple, smooth and its function is:

$$h_j = \exp\left(-\frac{\|X - C_j\|^2}{2b_j^2}\right) \quad (j=1, 2, 3, \dots, m) \quad (1)$$

The output of the RBF network is:

$$y_m(k) = \sum_{j=1}^m w_j(k) h_j(k) \quad (2)$$

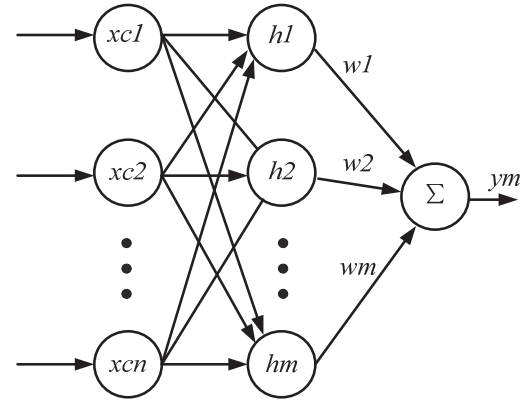


Fig 3: The topology of RBF neural network

Where, $C_j = [c_{j1}, c_{j2}, \dots, c_{jm}]^T$ in Formula (4) is the center of the j -th hidden layer node, $b_j = [b_1, b_2, \dots, b_m]^T$ is the width of j -th hidden layer node, $W = [w_1, w_2, \dots, w_m]^T$ is the connection weight of the hidden layer node and output space.

In this article the learning method of RBF network is gradient descent algorithm^[9] which principle is adjusting the center vector, width vector and weight of the hidden layer according to the requirements of minimize objective function, the objective function is:

$$\begin{cases} e_m(k) = y(k) - y_m(k) \\ J = \frac{1}{2} e_m(k)^2 \end{cases} \quad (3)$$

The center vector, width vector and weight are updating as follows:

$$\begin{cases} \Delta w_j = -\frac{\partial J}{\partial w_j} = e_m(k) \frac{\partial y_m(k)}{\partial w_j} = e_m(k) h_j \\ w_j(k) = w_j(k-1) + \eta \Delta w_j + \alpha(w_j(k-1) - w_j(k-2)) \\ \quad + \beta(w_j(k-1) - w_j(k-2)) \\ b_j(k) = b_j(k-1) + \eta w_j \Delta w_j \frac{\|X_c - C_j\|^2}{b_j^3} + \\ \quad \alpha(b_j(k-1) - b_j(k-2)) + \beta(b_j(k-1) - b_j(k-2)) \\ c_{ji}(k) = c_{ji}(k-1) + \eta w_j \Delta w_j \frac{xc_i - c_{ji}}{b_j^2} + \\ \quad \alpha(c_{ji}(k-1) - c_{ji}(k-2)) + \beta(c_{ji}(k-1) - c_{ji}(k-2)) \end{cases} \quad (4)$$

Where, η in Formula (7) is the learning rate, $\alpha\beta$ is the momentum factor, definition to performance indicators of identification network is:

$$\frac{\partial y(k)}{\partial u(k)} \approx \frac{\partial y_m(k)}{\partial u(k)} \quad (5)$$

Set the input of RBF network is $xc_1 = u(k)$, so have:

$$\begin{aligned} \frac{\partial y_m(k)}{\partial u(k)} &= \frac{\partial y_m(k)}{\partial xc_1} = \sum_{j=1}^m \frac{\partial y_m(k)}{\partial h_j} \frac{\partial h_j}{\partial xc_1} \\ &= \sum_{j=1}^m w_j h_j \frac{c_{j1} - xc_1}{b_j^2} \end{aligned} \quad (6)$$

By comparing the above formulas, RBF neural network is a network with local approximation, it approaches global approximation by achieving local approximation in each hidden layer unit, which can speed up the learning process of single neuron PID controller and reduce the error to improve control accuracy.

2.2 The fuzzy control algorithm

The single neuron PID controller does not have the ability of self-tuning gain K . K is the most sensitive parameter of the entire controller, which determines both the response speed and the overshoot size. The larger the value of K is, the faster the controller is response but the excessive amount of K value will cause the overshoot, so that the K value at the beginning of the control process should be taken for a larger value to increase the response speed and increased when deviation is growing. On the other hand, the value of K should be reduced to avoid an increase overshoot when the controller is in a stable state.

In this article, fuzzy controller is set by dual-input (error e and error change rate ec), single-output (dk , the variation of sensitive coefficient K). Scale factors are ke , kec and kcd . By setting different scale factor, it can correspond to the different input and output ranges. Sensitivity coefficient K value is corrected by an incremental approach. If K_0 is the initial K , then $K = K_0 + dk * kd$ ($k \neq 0$)

Set e , ec 's domain as $[-3, 3]$ and their five fuzzy subset are: NB (negative big), NS (negative small), Z (zero), PS (positive small), PB (positive big), their membership functions are shown in Figure 4; dk 's domain is $[-0.06, 0.06]$; three fuzzy subsets are N (negative), Z (zero) and P (positive) corresponding to the value of K which should be increased, decreased and remain stable, dk 's membership function is shown in Figure 5.

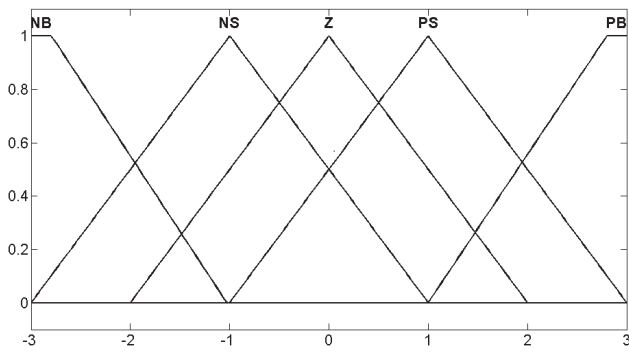


Fig 4: The membership function of e and ec

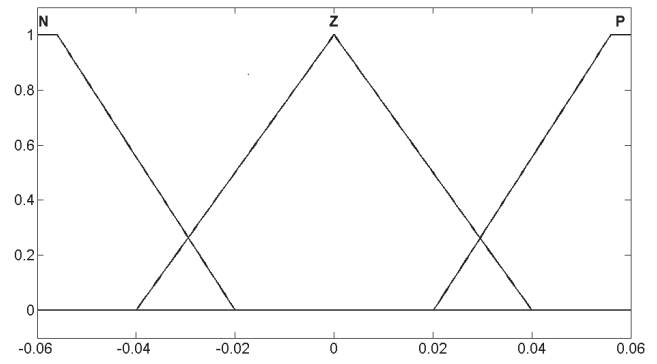


Fig 5: The membership function of dk

The core of the fuzzy controller is fuzzy inference process, the purpose of gain adaptive is increase K values when e and ec is larger and reduced it when the e and ec is small. Table 1 can be drawn by controlling demand and expertise, where the process of using Mamdani fuzzy inference algorithm^[10] (weighted average fuzzy). The following fuzzy inference obtains relations through data in Table 1.

Rule: If e is A and ec is B then dk is C

Table 1: Table fuzzy inference rule

e	ec				
	NB	NS	Z	PS	PB
NB	N	N	N	N	Z
NS	N	N	N	Z	P
Z	N	N	Z	P	P
PS	N	Z	P	P	P
PB	Z	P	P	P	P

2.3 Improved control system algorithm

In this article, a single neuron PID control algorithm combines the RBF neural networks and fuzzy control had been studied. Summarizes the controller algorithm as following:

① Gives initial weights, learning rate of the single neuron PID controller; Set RBF network parameters: the output weights vector W , center vector C_j and width vector b_j ; Gives initial fuzzy controller parameters: e , ec , ke , kec and kcd .

② Calculates error by incremental algorithm; Calculates $e(k)$, $ec(k)$ and gain coefficient K ; Calculates the output of single neuron PID controller's $u(k)$ by the formula (7).

$$\begin{cases} w' = \sum_{i=1}^3 |w_i(k)| \\ u(k) = u(k-1) + K \sum_{i=1}^3 w_i(k) x_i(k) / w' \end{cases} \quad (7)$$

$$e(k) = rin(k) - yout(k) \quad (8)$$

$$\begin{cases} x_1(k) = e(k) - e(k-1) \\ x_2(k) = e(k) \\ x_3(k) = e(k) - 2e(k-1) + e(k+2) \end{cases} \quad (9)$$

③ Transports $u(k)$ to RBF network and the controlled object to produce the identified output and actual output of the next step.

④ The output of RBF network is calculated by the formula (2).

⑤ Use the value of actual output $yout(k+1)$ minus RBF network output $(k+1)$ to fixed the RBF network. Then calculated central vector, the width vector and the connection weights by formula (3) and (4).

⑥ The performance index function of single neuron weights:

$$e(k+1) = \frac{1}{2} (r_{in}(k+1) - yout(k+1))^2 \quad (10)$$

The weight at the next step is:

$$wi(k+1) = wi(k) - \eta i \frac{\partial e(k+1)}{\partial wi(k)} \quad (11)$$

Then:

$$\frac{\partial e(k+1)}{\partial wi(k)} = -e(k+1) \frac{\partial yout(k+1)}{\partial u(k)} \cdot \frac{\partial u(k)}{\partial wi(k)} \quad (12)$$

Calculates $\frac{\partial u(k)}{\partial wi(k)}$ by formula (7), so have:

$$\frac{\partial e(k+1)}{\partial wi(k)} = -e(k+1) \frac{\partial yout(k+1)}{\partial u(k)} x_i(k) \quad (13)$$

Because y_{out} approaching $yout$ during learning process so that $\frac{\partial yout(k+1)}{\partial u(k)}$ can be calculated by formula (5), and final weight correction formula of single neuron is:

$$w_i(k+1) = w_i(k) + \eta_i e(k+1) x_i \sum_{j=1}^m w_j h_j \frac{c_{j1} - x_2}{b_j^2} \quad (14)$$

⑦ Set $k = k + 1$, return to step ②

3 Simulation and Analysis

In order to verify the performance of composite control system the Simulink software package has been used in this article. Because the RBF neural network and single neuron PID controller can not be directly modify so a S-function has been used here, S-function is written by Matlab language so it can easily define and embedded it into the simulation model^[11].

Fuzzy controller uses Simulink Fuzzy Toolbox, determine the number of input and output, set the parameters and set the membership function according to the design.

Firstly, an inertia plus time delay model has used as controlled object here which can represent most industrial

controlled object^[12]. So it can be use to detect control system dynamic performance, robustness and adaptability, the controlled object transfer function is:

$$G(s) = \frac{2.3e^{-0.4s}}{3.05s + 1} \quad (16)$$

Simulation input step signal amplitude is 1. 10% of sudden disturbance is added when $t=150s$. Whole set of conventional PID parameters are $K_p = 0.3$, $K_i = 0.35$ and $K_d = 0.1$. The same parameters are used in other controllers to compare the difference between the proposed control method, the single neuron PID controller and the conventional PID. Set the same learning rate of proportion, integral and differential in compound controller and single neuron PID controller with 0.8, 0.52 and 0.2. Set the same initial K_0 of Fuzzy controller and K of single neuron PID controller with 0.01; Composite controller other initial parameters are:

RBF neural network has 2 units of input layer, 6 units of hidden layer and one unit of output layer. RBF neural network learning rate η is 0.2; Momentum factor α is 0.05, β is 0.02; Center vector C_i is $10 * \text{ones}(2, 6)$; Width vector b_i is $50 * \text{ones}(6, 1)$, the weight vector w is $10 * \text{ones}(6, 1)$. Fuzzy controller input scale factor k_e , k_{ec} are 1.5 and 25 and the output scale factor k_{kd} is 1×10^{-3} . The step response of the system is shown in Figure 6. Figure 7 is the recognized Jacobian information curve; figure 8 is the weights adjustment curve of control system.

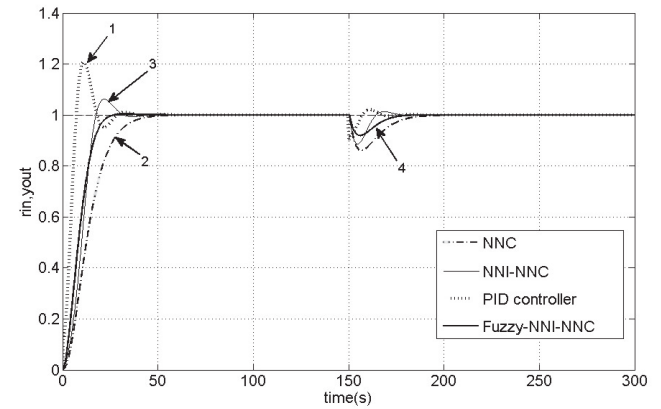


Fig6: The step response curve

Curve 1 in Figure 6 is the conventional PID response curve. Curve 2 is the single neuron PID (NNC) response curve. Curve 3 is the single neuron PID with RBF identification (NNI-NNC) response curve. Curve 4 is the proposed control algorithm curve (Fuzzy-NNI-NNC) in this article. From the figure, the existence of conventional PID curve shocks larger overshoot can be seen. Although there is no overshoot and oscillation in single neuron PID but the long weights self-tuning process cause a lag. Because the value of K can not self-tune, there was a certain overshoot at curve 3.

Comparing the above control method, fuzzy value adjustments through complex RBF neural networks in this article reduce overshoot remarkable and get a better dynamic performance, robustness and immunity.

Comparing through step response curves in Figure 6, an obvious conclusion can be found that K value is very important to the single neuron controller. The only use of RBF network improves the single neuron PID may get some overshoot when improper K value set is elicited. Figure 9 shows a sensitivity adjustment factor K. The value K can be seen to increase the learning speed increases at the beginning and reduced to suppress overshoot when system stable.

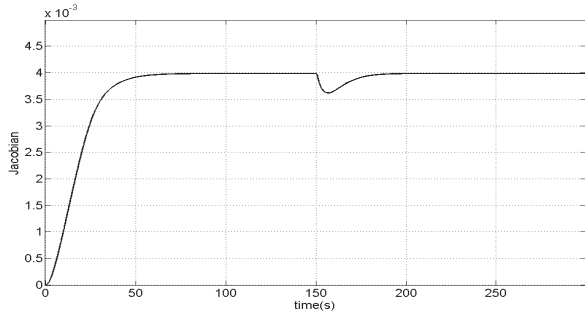


Fig7: Identification of Jacobian information

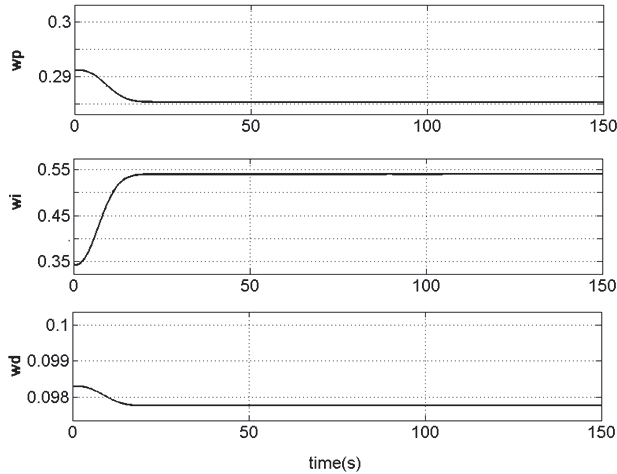


Fig8: The system weight online adjustment curve

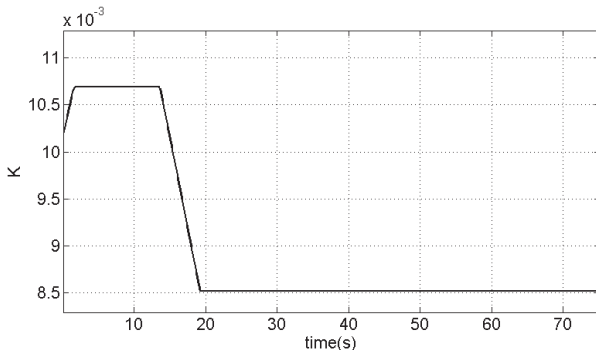


Figure 9: K value online adjustment curve

In order to study the effect of K value online correction on tracking performance and dynamic performance of control

systems, the article selects second continuous $\frac{s+1}{s^2+3s+1}$ as controlled object, used the conventional PID controller, the proposed composite controller and single neuron PID controller with RBF identification to track sinusoidal signal, the input signal frequency is 1 (rad/sec), signal amplitude is 1, and figure 10 is the response curve.

Curve 1 in Figure 10 is the response curve of proposed algorithm; Curve 2 is the response curve of single neuron PID controller with RBF identification; Curve 3 is the response curve of the conventional PID controller. By comparing the curves in Figure 10 an obvious conclusion can be found that proposed algorithm can track the sin signal accurately without overshoot in a short time, but the single neuron PID controller with RBF identification has a certain amount of overshoot at the pole position because the K value can not online tuning, what is more, the traditional PID controller has both residual error and lag.

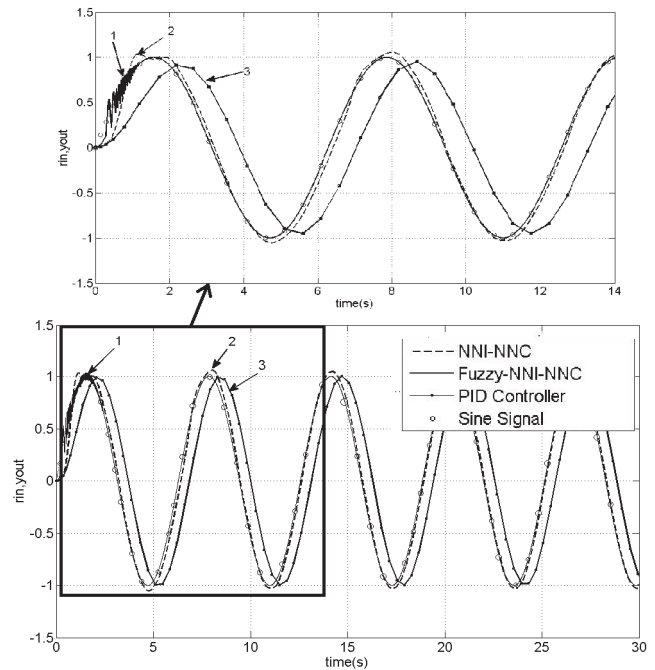


Fig10: Sine signal tracking curve

Figure 11 is the adjustment curve of K value. The K value is adjusting synchronously with the input sinusoidal signal. Whenever the value of the tracking is near the curve, the poles of K will be reduced to weaken overshoot. By adjusting the value of K, the controller can better track the sinusoidal signal. The simulation results show that the K value with online tuning make controller has better tracking ability, good dynamic performance and robustness.

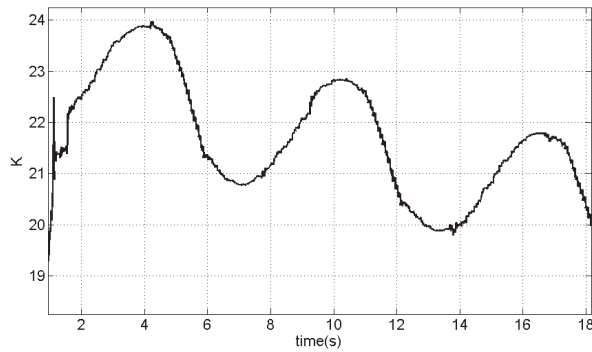


Fig 11: K value online adjustment curve

4 Conclusion

In order to overcome the low learning speed and gain can not auto-tuning of single neural network PID controller, a new approach had been studied in this article which use fuzzy control and RBF network identification to improve the single neuron PID controller, after simulated by Simulink some conclusions can be summarized as follows:

- (1) Complex network PID control method has a better control effect on nonlinear and time-delay controlled objects. It is significantly better than the conventional PID controller and the controller improved only by single neuron.
- (2) Improved control algorithm has better tracking performance and control accuracy. It not only retains the advantages of the single neuron PID and RBF neural network through combining them together, but also solves their deficiency of not being able to self-tuning gain through fuzzy control. This method has a guiding role for the intelligent design of the future controllers.

REFERENCES

- [1] Fang, Hong Yu, Xiaohong, Design and Simulation of Neuron PID Controller, International Conference on Information Technology & Computer Engineering and Management Sciences[C], 2011.
- [2] Jie Xiu, Shi Yu-wang, Yan Xiu, Fuzzy adaptive single neuron NN control of brushless DC motor, Neural computing & applications[A], Vol.22, No.3, 1045-1047, 2013
- [3] Yang shuai, Ren Jin-xia, A kind of improved single neuron PID controller, Manufacturing Automation [A], Vol.32, No.1, 119-121, 2010.
- [4] 刘金琨, 先进PID控制及其Matlab仿真, 北京电子工业出版社, 2003.
- [5] Hassan B, Kazemian. Development of an Intelligent Fuzzy Controller. IEEE Fuzzy system conference[C], 517-520, 2001.
- [6] Wu Bao-gang, Ying Hao, Review of Fuzzy PID Control Techniques and Some Important Issues, Acta Automatica Sinica, Vol.27, No.4, 568-570, 2001.
- [7] Jiang Ying, Wang Jing, Improved Single Nerve Cell PID Control Based on RBF, Chinese Journal of Scientific Instrument, Vol.26, 2005.
- [8] Cheng Cheng, Min-Sen CHIU, Adaptive Single-Neuron Controller Design for Nonlinear Process Control. Journal of Chemical Engineering of Japan[J], Vol.41, No.8, 2008
- [9] Miao Zhuo-guang, He Xiu-ran, Multivariable Decoupling Control Based on Fuzzy RBF Network for Aeroengine,

Journal of Air Force Engineering University[J], Vol.10, No.2, 10-13, 2009.

- [10] Zhang Shi-Tao, Yang Feng, Research and Simulation of Single Neuron PID Controller, Mechanical Engineering & Automation[J], Vol.154, No.3, 71-73, 2009.
- [11] Xu Bing, Cheng Xu-de, Wang Hong-li, 基于Matlab的Mamdani与Sugeno型模糊推理研究, JOURNAL OF COMPUTER APPLICATIONS[J], Vol.26, 223-225, 2006.
- [12] 薛定宇, 控制系统仿真与计算机辅助设计, 北京机械工业出版社[M], 2005.
- [13] Zhao Bin, You Wen, The Research of a Model Algorithm Control on the One-order Inertia Process with Pure-time Delay, Automation in Petro-Chemical Industry[J], Vol.44, No.1, 44-46, 2008.
- [14] Wang Xu-guang, SUN Qu, A Neuron PID Speed Regulator Based on RBF Identification, Techniques of Automation and Application[J], Vol.27, No.6, 57-60, 2008.
- [15] Ding Jun, Xu Yong-mao, Single Neuron Adaptive PID Controller and its Applications, Control Engineering of China[J], Vol.11, No.1, 27-29, 2004.
- [16] Xia Chang-liang, WANG Ming-chao, Single Neuron PID Control For Switched Reluctance Motors Based On RBF Neural Network, Proceedings of the CSEE[J], Vol.25, No.15, 163-164, 2005.